

VANAVARA, Russian Federation

WMO: 249080

Lat: 60.352N

Lon: 102.301E

Elev: 854

StdP: 14.25

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-55.8	-50.7	-59.7	0.2	-55.7	-54.9	0.2	-50.7	11.3	-3.3	9.8	1.3	0.6	50	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	85.2	64.3	81.3	62.8	77.6	61.0	66.5	80.1	64.7	76.5	63.0	74.1	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	86.8	69.3	60.4	80.9	68.5	58.5	75.6	67.1	31.7	80.4	30.3	76.4	29.0	74.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.6	8.2	DB	-58.8	90.9	5.6	3.0	-62.9	93.1	-66.2	94.8	-69.4	96.5	-73.4	98.7	
			WB	-58.5	69.5	5.7	2.1	-62.6	71.0	-65.9	72.2	-69.1	73.4	-73.2	74.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	-18.5	-11.9	9.7	28.5	42.8	59.6	63.1	56.9	42.5	25.2	-2.1	-18.6
	DBStd	32.24	17.53	16.53	14.68	10.59	8.37	7.69	5.97	6.62	7.60	11.66	19.54	20.62
	HDD50	10741	2123	1732	1248	645	255	13	1	17	249	770	1562	2126
	HDD65	15319	2588	2152	1713	1095	689	195	109	264	676	1235	2012	2591
	CDD50	990	0	0	0	0	31	300	406	230	23	0	0	0
	CDD65	93	0	0	0	0	1	32	49	11	0	0	0	0
	CDH74	1593	0	0	0	0	20	649	701	217	6	0	0	0
	CDH80	439	0	0	0	0	1	198	184	56	0	0	0	0
	WSAvg	3.4	2.5	2.8	3.6	4.2	4.3	3.8	3.6	3.3	3.4	3.6	3.2	2.5
	PrecAvg	15.9	0.8	0.5	0.6	1.0	1.4	1.9	2.1	2.4	1.6	1.3	1.3	1.0
Precipitation	PrecMax	23.3	1.2	1.6	1.1	3.1	2.5	4.3	5.2	6.0	4.8	2.4	2.6	1.8
	PrecMin	10.7	0.3	0.1	0.0	0.4	0.3	0.3	0.1	0.3	0.2	0.6	0.6	0.4
	PrecStd	3.1	0.2	0.3	0.2	0.6	0.5	0.8	1.3	1.3	0.9	0.5	0.4	0.3
	PrecStd	3.1	0.2	0.3	0.2	0.6	0.5	0.8	1.3	1.3	0.9	0.5	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	29.5	46.8	60.4	77.6	89.9	89.1	86.0	73.4	53.1	37.5	28.0
		MCWB	21.6	27.3	36.4	44.0	56.6	64.8	66.9	65.8	58.5	45.4	35.1	27.0
	2%	DB	18.7	25.8	40.4	55.0	70.3	85.7	84.9	79.6	65.7	47.1	34.0	23.0
		MCWB	17.5	23.6	32.8	42.4	52.5	63.7	64.7	62.5	53.8	42.0	32.4	22.4
	5%	DB	14.4	20.5	35.8	50.1	65.2	81.6	81.5	74.9	60.3	42.9	31.4	18.4
		MCWB	13.7	18.8	30.3	39.3	49.6	62.1	63.4	60.2	51.6	39.5	29.7	17.7
	10%	DB	9.0	14.3	32.0	45.4	59.9	77.4	78.0	70.9	55.6	39.2	27.6	13.0
		MCWB	8.2	13.2	27.7	36.6	46.9	59.7	61.9	58.4	49.6	36.6	26.2	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	27.5	37.3	45.5	58.2	68.1	69.0	67.4	59.4	46.4	35.5	27.2
		MCDB	22.6	28.8	44.7	56.5	73.6	84.0	81.5	81.6	68.9	50.4	36.6	28.0
	2%	WB	17.6	23.7	33.4	42.8	54.4	65.5	67.1	64.5	55.6	42.7	32.6	22.3
		MCDB	18.7	25.7	39.2	53.7	66.6	80.9	80.2	75.4	62.7	46.6	33.9	22.9
	5%	WB	13.5	18.8	31.0	40.1	51.2	63.5	65.4	62.3	52.8	39.8	29.7	17.7
		MCDB	14.3	20.7	35.7	49.0	62.8	77.7	76.8	72.1	58.7	42.6	31.2	18.5
	10%	WB	8.3	13.2	27.5	37.4	47.9	61.7	63.9	60.1	50.1	36.8	26.2	12.5
		MCDB	9.0	14.3	31.8	44.5	59.4	74.2	74.1	67.9	55.2	39.0	27.7	13.1
Mean Daily Temperature Range	5% DB	MDBR	18.6	26.1	29.3	22.0	20.5	24.1	22.5	21.2	17.6	13.9	18.3	17.0
		MCDBR	19.1	18.6	27.2	28.1	31.4	33.3	30.4	30.1	27.6	18.1	12.4	16.3
		MCWBR	18.6	17.4	21.0	17.0	16.6	13.3	12.4	14.8	16.3	13.3	11.8	16.1
	5% WB	MCDBR	19.0	18.6	25.3	25.3	28.4	28.6	24.3	24.6	22.6	16.8	12.6	15.9
		MCWBR	18.5	17.5	20.0	16.1	16.0	12.0	10.9	13.2	14.2	12.8	12.3	15.8
		MCWBR	18.5	17.5	20.0	16.1	16.0	12.0	10.9	13.2	14.2	12.8	12.3	15.8
Clear-Sky Solar Irradiance	taub		0.255	0.289	0.315	0.348	0.369	0.396	0.462	0.432	0.341	0.308	0.253	0.224
	taud		1.955	2.010	2.047	2.093	2.302	2.339	2.125	2.170	2.433	2.282	2.003	1.900
	Ebn at Noon		146	213	251	266	271	265	240	237	248	212	146	107
	Edh at Noon		15	27	36	41	36	36	43	38	25	20	14	10
All-Sky Solar Radiation	RadAvg		148	454	952	1448	1636	1819	1656	1267	763	351	173	81
	RadStd		10	30	56	95	151	138	130	104	89	37	21	9

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page